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(Autonomous Institute, Affiliated to VTU)
Department of Computer Science and Engineering

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Inference Rules for FDs

Given a set of FDs F , we can **infer** additional FDs that hold whenever the FDs in F hold

Armstrong's inference rules:

- IR1. (**Reflexive**) If Y *subset-of* X , then $X \rightarrow Y$
- IR2. (**Augmentation**) If $X \rightarrow Y$, then $XZ \rightarrow YZ$
 - (Notation: XZ stands for $X \cup Z$)
- IR3. (**Transitive**) If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$

IR1, IR2, IR3 form a **sound** and **complete** set of inference rules

- These are rules that hold and all other rules that hold can be deduced from these

Inference Rules for FDs

Some additional inference rules that are useful:

- **Decomposition:** If $X \rightarrow YZ$, then $X \rightarrow Y$ and $X \rightarrow Z$
- **Union:** If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$
- **Pseudotransitivity:** If $X \rightarrow Y$ and $WY \rightarrow Z$, then $WX \rightarrow Z$

The last three inference rules, as well as any other inference rules, can be deduced from IR1, IR2, and IR3 (completeness property)

Proof of Inference Rules

Proof of IR1. Suppose that $X \supseteq Y$ and that two tuples t_1 and t_2 exist in some relation instance r of R such that $t_1[X] = t_2[X]$. Then $t_1[Y] = t_2[Y]$ because $X \supseteq Y$; hence, $X \rightarrow Y$ must hold in r .

Proof of IR2 (by contradiction). Assume that $X \rightarrow Y$ holds in a relation instance r of R but that $XZ \rightarrow YZ$ does not hold. Then there must exist two tuples t_1 and t_2 in r such that (1) $t_1[X] = t_2[X]$, (2) $t_1[Y] = t_2[Y]$, (3) $t_1[XZ] = t_2[XZ]$, and (4) $t_1[YZ] \neq t_2[YZ]$. This is not possible because from (1) and (3) we deduce (5) $t_1[Z] = t_2[Z]$, and from (2) and (5) we deduce (6) $t_1[YZ] = t_2[YZ]$, contradicting (4).

Proof of IR3. Assume that (1) $X \rightarrow Y$ and (2) $Y \rightarrow Z$ both hold in a relation r . Then for any two tuples t_1 and t_2 in r such that $t_1[X] = t_2[X]$, we must have (3) $t_1[Y] = t_2[Y]$, from assumption (1); hence we must also have (4) $t_1[Z] = t_2[Z]$ from (3) and assumption (2); thus $X \rightarrow Z$ must hold in r .

Proof of Inference Rules

Proof of IR4 (Using IR1 through IR3).

1. $X \rightarrow YZ$ (given).
2. $YZ \rightarrow Y$ (using IR1 and knowing that $YZ \supseteq Y$).
3. $X \rightarrow Y$ (using IR3 on 1 and 2).

Proof of IR5 (using IR1 through IR3).

1. $X \rightarrow Y$ (given).
2. $X \rightarrow Z$ (given).
3. $X \rightarrow XY$ (using IR2 on 1 by augmenting with X ; notice that $XX = X$).
4. $XY \rightarrow YZ$ (using IR2 on 2 by augmenting with Y).
5. $X \rightarrow YZ$ (using IR3 on 3 and 4).

Proof of IR6 (using IR1 through IR3).

1. $X \rightarrow Y$ (given).
2. $WY \rightarrow Z$ (given).
3. $WX \rightarrow WY$ (using IR2 on 1 by augmenting with W).
4. $WX \rightarrow Z$ (using IR3 on 3 and 2).

Problem on Inference Rules:

1. Prove or disprove the following inference rules for functional dependencies. A proof can be made by using inference rules R1 through R3. A disproof should be performed by demonstrating a relation that satisfies the conditions and functional dependencies in the left-hand side of the inference rule but does not satisfy the dependencies in the right-hand side.

Prove:

- a. $\{W \Rightarrow Y, X \Rightarrow Z\} \models WX \Rightarrow Y$
- b. $\{X \Rightarrow Y, Z \text{ is a subset of } Y\} \models X \Rightarrow Z$
- c. $\{X \Rightarrow Y, X \Rightarrow W, WY \Rightarrow Z\} \models X \Rightarrow Z$
- d. $\{X \Rightarrow Y, XY \Rightarrow Z\} \models X \Rightarrow Z$
- e. $\{X \Rightarrow Y, Z \Rightarrow W\} \models XZ \Rightarrow YZ$
- f. $\{X \Rightarrow Y, Y \Rightarrow Z\} \models X \Rightarrow YZ$

Disprove:

- g. $\{XY \Rightarrow Z, Y \Rightarrow W\} \models XW \Rightarrow Z$

Closure of a set F of FDs

- **Closure** of a set F of FDs is the set F^+ of all FDs that can be inferred from F .
- **Closure** of a set of attributes X with respect to F is the set X^+ of all attributes that are functionally determined by X . X^+ can be calculated by repeatedly applying IR1, IR2, IR3 using the FDs in F .

Algorithm 10.1: Determining X^+ , the Closure of X under F

$X^+ := X$;

repeat

$\text{old}X^+ := X^+$;

 for each functional dependency $Y \rightarrow Z$ in F do

 if $X^+ \supseteq Y$ then $X^+ := X^+ \cup Z$;

until $(X^+ = \text{old}X^+)$;

The following set F of functional dependencies that should hold on EMP_PROJ;

$$F = \{SSN \rightarrow ENAME, \\ PNUMBER \rightarrow \{PNAME, PLOCATION\}, \\ \{SSN, PNUMBER\} \rightarrow HOURS\}$$

we calculate the following closure sets with respect to F;

$$\{SSN\}^+ = \{SSN, ENAME\}$$

$$\{PNUMBER\}^+ = \{PNUMBER, PNAME, PLOCATION\}$$

$$\{SSN, PNUMBER\}^+ = \{SSN, PNUMBER, ENAME, PNAME, PLOCATION, HOURS\}$$

Problem:

Consider the following relation schema and set of functional dependencies:
Emp-Dept (SIN, E_Name, B_Date, Address, D_Num, D_Name, D_Manager)
 $F = \{ \text{SIN} \rightarrow \{ \text{E_Name}, \text{B_Date}, \text{Address}, \text{D_Num} \},$
 $\text{D_Num} \rightarrow \{ \text{D_Name}, \text{D_Manager} \}$
 $\}$

Calculate the closure of $\{\text{SIN}\}^+$ and $\{\text{D_Num}\}^+$ with respect to F.

R(A,B,C,D,E,F)

FDs S = $\{ \text{AB} \rightarrow \text{C}, \text{BC} \rightarrow \text{AD}, \text{D} \rightarrow \text{E}, \text{CF} \rightarrow \text{B} \}$

What is the closure of:

$\{ \text{A}, \text{B} \}^+$

Equivalence of Sets of Functional Dependencies

Two sets of FDs F and G are **equivalent** if:

- Every FD in F can be inferred from G , and
- Every FD in G can be inferred from F
- Hence, F and G are equivalent if $F^+ = G^+$

Definition (**Covers**):

- F **covers** G if every FD in G can be inferred from F
 - (i.e., if $G^+ \text{ subset-of } F^+$)

F and G are equivalent if F covers G and G covers F .

Equivalence of Sets of Functional Dependencies

Definition. A set of functional dependencies F is said to **cover** another set of functional dependencies E if every FD in E is also in F^+ ; that is, if every dependency in E can be inferred from F ; alternatively, we can say that E is **covered** by F .

Definition. Two sets of functional dependencies E and F are **equivalent** if $E^+ = F^+$. Hence, equivalence means that every FD in E can be inferred from F , and every FD in F can be inferred from E ; that is, E is equivalent to F if both the conditions E covers F and F covers E hold.

Examples:

$R=(A,B,C,D,E,F)$

$F1=\{A \rightarrow BC, B \rightarrow CDE, AE \rightarrow F\}$

$F2=\{A \rightarrow BCF, B \rightarrow DE, E \rightarrow AB\}$

Check whether F1 and F2 are equivalent or not.

Solution

To check F1 covers F2 –

$A^+=\{A,B,C,D,E,F\}$ contains B,C,F

$B^+=\{B,C,D,E\}$ contains D,E

$E^+=\{E\}$ contains A,B

So F1 does not cover F2.

Hence F1 and F2 are not equivalent.

Consider another example where two functional dependencies are equivalent.

$R=(A,C,D,E,H)$

$F1=\{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\},$

$F2=\{A \rightarrow CD, E \rightarrow AH\}$

Check whether F1 and F2 are equivalent or not?

Solution

To check F1 covers F2 –

$A^+=\{A,C,D\}$ contains C,D

$E^+=\{A,D,E,H\}$ contains A,H

So F1 covers F2

To check F2 covers F1:

$A^+=\{A,C,D\}$ contains C

$\{A,C\}^+=\{A,C,D\}$ contains D

$E^+=\{A,C,D,E,H\}$ contains A,D,H

So F2 covers F1.

Hence F1 and F2 are equivalent.

Minimal Sets of FDs

A set of FDs is **minimal** if it satisfies the following conditions:

1. Every dependency in F has a single attribute for its RHS.
2. We cannot replace any dependency $X \rightarrow A$ in F with a dependency $Y \rightarrow A$, where Y proper-subset-of X ($Y \subset X$) and still have a set of dependencies that is equivalent to F .
3. We cannot remove any dependency from F and have a set of dependencies that is equivalent to F .

Minimal Sets of Functional Dependencies

Algorithm 10.2: Finding a Minimal Cover F for a Set of Functional Dependencies E

1. Set $F := E$.
2. Replace each functional dependency $X \rightarrow \{A_1, A_2, \dots, A_n\}$ in F by the n functional dependencies $X \rightarrow A_1, X \rightarrow A_2, \dots, X \rightarrow A_n$.
3. For each functional dependency $X \rightarrow A$ in F
 for each attribute B that is an element of X
 if $\{ F - \{X \rightarrow A\} \} \cup \{(X - \{B\}) \rightarrow A\}$ is equivalent to F ,
 then replace $X \rightarrow A$ with $(X - \{B\}) \rightarrow A$ in F .
4. For each remaining functional dependency $X \rightarrow A$ in F
 if $\{ F - \{X \rightarrow A\} \}$ is equivalent to F ,
 then remove $X \rightarrow A$ from F .

In Chapter 11 we will see how relations can be synthesized from a given set of dependencies E by first finding the minimal cover F for E .

Minimal Sets of FDs

- Every set of FDs has an equivalent minimal set.
- There can be several equivalent minimal sets.
- There is no simple algorithm for computing a minimal set of FDs that is equivalent to a set F of FDs.
- To synthesize a set of relations, we assume that we start with a set of dependencies that is a minimal set.

Find Minimal Cover/ Canonical Cover of E

Let the given set of FDs be $E : \{B \rightarrow A, D \rightarrow A, AB \rightarrow D\}$. We have to find the minimum cover of E.

- All above dependencies are in canonical form; so we have completed step 1 of Algorithm 10.2 and can proceed to step 2.

In step 2 we need to determine if $AB \rightarrow D$ has any redundant attribute on the left-hand side; that is, can it be replaced by $B \rightarrow D$ or $A \rightarrow D$?

- Since $B \rightarrow A$, by augmenting with B on both sides (IR2), we have $BB \rightarrow AB$, or $B \rightarrow AB$

(i). However, $AB \rightarrow D$ as given (ii).

- Hence by the transitive rule (IR3), we get from (i) and (ii), $B \rightarrow D$. Hence $AB \rightarrow D$ may be replaced by $B \rightarrow D$.

- We now have a set equivalent to original E, say $E' : \{B \rightarrow A, D \rightarrow A, B \rightarrow D\}$. No further reduction is possible in step 2 since all FDs have a single attribute on the left-hand side.

- In step 3 we look for a redundant FD in E' . By using the transitive rule on $B \rightarrow D$ and $D \rightarrow A$, we derive $B \rightarrow A$. Hence $B \rightarrow A$ is redundant in E' and can be eliminated.

- Hence the minimum cover of E is $\{B \rightarrow D, D \rightarrow A\}$

Thank you